The current elementary VSL stack

Consolidated equations, derivations and conditional results

Feedback 8 mathematical report — 16 October 2026

Purpose. Assemble the selected model and the mathematical work supporting its improvements in one notation. This is a consolidation of the S13 reference, H1/T1, selected S-E-P/R-E/O-R/W-F, the full-domain forward argument at WF-Q1 and the physical-identifiability review at MP6-Q1. It is not a new model selection, experiment, derivative validation or execution plan. No rejected formula is reinstated as current behavior.

The central chain. Initial footprint and common advice determine receiving-limited preparation. Boundary histories transport the resulting streams into a finite terminal. A/E service combines present maneuver work with retained effects of earlier E entries. Conservation and complete departures determine local total time. The proofs qualify this chain under declared restrictions, not all lane-drop traffic.

Reading guide. Sections 1–5 give the original reference, geometry, upstream law and terminal interfaces. Sections 6–8 specify present service and recovery. Sections 9–11 give causality, positive service, event continuation, clearance and accounting. Section 12 records information and sensitivity limits. Section 13 reproduces the selected analytical coexistence construction. Sections 14–15 collect physical-identification deductions and the source/status map. These supporting proofs are available for human review; symbolic checks of identities are not machine certification of quantified physical or hybrid-system claims.

Status. The stack is functionally complete as a conditional mathematical model. Applied MP6, amended action/gradient and SUMO evidence contracts remain open. Step 3 is unfinished; Step 4 remains unopened. S13's closed certificate remains valid for S13 and is not transferred to the amended model. "M2" in the review is a mechanism label, not project implementation Milestone 2. The urban task and ramp line remain closed.

Companion: plain-language old-to-new model recap

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1 The S13 backbone and its original closed calculation

S13 starts with total continuous mass M and ending fraction p_s in preparation state s. The other mass is a symmetric continuing group. A common constant advice v gives traversal time $\tau = L_{13}/v$. With constant positive completion rate λ_{13},

$$\dot{E} = -\lambda_{13}E, \quad \dot{C} = \lambda_{13}E, \qquad E(\tau) = p_s M e^{-\lambda_{13}\tau}. \tag{1}$$

Target receiving is assumed nonbinding during preparation. Remaining E consumes $\alpha > 1$ receiving work per mass at a fixed work rate μ:

$$W_{13} = M + (\alpha - 1)E(\tau), \quad T_{13} = W_{13}/\mu, \quad \overline{J}_{13} = \tau + T_{13}/2 = \tau + \frac{M}{2\mu} + K_{13}p_s e^{-\lambda_{13}\tau}, \tag{2}$$

where $K_{13} = (\alpha - 1)M/(2\mu)$. The half-clearance term follows from homogeneous proportional fluid-batch departures. It is not a general formula for an arbitrary arrival or departure curve.

The original action result follows from

$$\overline{J}_{13,\tau} = 1 - K_{13}\lambda_{13}p_s e^{-\lambda_{13}\tau}, \qquad \overline{J}_{13,\tau\tau} = K_{13}\lambda_{13}^2 p_s e^{-\lambda_{13}\tau} > 0. \tag{3}$$

If two aggregate-history-matched preparations satisfy

$$K_{13}\lambda_{13}p_L < e^{\lambda_{13}\tau_{\min}} < K_{13}\lambda_{13}p_H < e^{\lambda_{13}\tau_{\max}}, \tag{4}$$

strict convexity gives fastest advice for L and an interior slower $\tau_H^* = \lambda_{13}^{-1}\log(K_{13}\lambda_{13}p_H)$ for H. The chain rule gives

$$\frac{d\overline{J}_{13}}{dv} = -\frac{L_{13}}{v^2}\left(1 - K_{13}\lambda_{13}p_s e^{-\lambda_{13}L_{13}/v}\right). \tag{5}$$

Identical aggregate observations force a deterministic observer-only controller to take the same action. The two unique minimizers differ; the maximum of their nonnegative continuous regret functions is positive on the compact action interval and has a positive minimum. This is S13's conditional information/action/analytical-gradient certificate.

The refinements below replace the constant completion, grouped receiving, batch timing and fixed-service reductions. Therefore (2)–(5) are historical reference results, not formulas for the current model. Their validity under S13's assumptions is preserved.

2 Current geometry, notation, units and admissible domain

2.1 Setting and initialized state

The road has preparation length L_p and terminal length $b > 0$, $L = L_p + b$. At decision time a uniform cohort occupies $[0, \ell]$, $0 < \ell \le L_p$, and the terminal is empty. E ends at L; A is its adjacent continuing lane and B the far continuing lane. Initial densities satisfy

$$e_0, a_0, b_0 \ge 0, \quad a_0 = b_0, \quad r = e_0 + a_0, \quad \rho_0 = e_0 + 2a_0, \quad M = \ell\rho_0 > 0. \tag{6}$$

There are no postdecision external arrivals. The constant common advice is $v \in [v_-, v_+]$, $0 < v_- < v_+$. It sets ideal upstream/free travel; queued boundary motion need not equal it. H1's remaining upstream occupied body is homogeneous. Terminal propagation is represented by boundary counts; there is no new interior spatial grid or independent approach-speed ODE.

Use

$$K = wk_j, \quad k_c(v) = \frac{K}{w+v}, \quad C(v) = \frac{Kv}{w+v}, \quad c(u) = \frac{Ku}{w+u}, \quad U(f) = \frac{wf}{K-f}, \quad U(C(v)) = v. \tag{7}$$

The shared A/E budget is $\mathcal{B}$; B as a lane index still means the far lane. The archived recovery reading p is written p_{rec} here. K_{13} in Section 1 is unrelated to $K = wk_j$.

2.2 Retained sufficient domain

All geometry and physical parameters are common across advice and observers. Retain H1/T1's initial/handoff conditions, in particular

$$\begin{aligned} &w, k_j, h, a_r > 0, \qquad \gamma \geq 1, \qquad e_0, a_0, b_0 \leq k_c(v_+), \\ &v_+ r < R \leq C(v_-), \qquad v_+ b_0 \leq R_B \leq C(v_-), \qquad M < k_j b. \end{aligned} \tag{8}$$

For immediate work set fixed reference units once,

$$\begin{aligned} &v_\star = v_+, \quad C_\star = C(v_\star), \quad c_0 = 1 + \delta, \quad \chi = \frac{\beta}{C_\star v_\star}, \\ &\delta, \beta \geq 0, \quad \eta = c_0 - \beta(1 + v_+/w) > 0, \quad c_{\max} = c_0 + \beta. \end{aligned} \tag{9}$$

The anchor fixes χ once; it is not recomputed when advice or a search box changes. The reference mass $m = 1$ vehicle-equivalent is fixed physical normalization, not packet size. Ordinary combined E/A arrival load fits R; any primary queue originates in additional forced work. These bounds are sufficient and may be conservative. They are not an empirically admitted operating range.

Symbols	Units and meaning
L, L_p, b, ℓ, h, z	m; lengths and traveled progress
$e, a, b_0, k_j, k_c, \rho_0$	veh-eq/m; densities
v, w, u_j, U, V	m/s; speeds or specified boundary speed quantities
$M, F_j, P_j, Y_j, N_j, q_j, m$	veh-eq; traffic mass, counts or reference mass
$f, g_j, \lambda_j, R, R_B, K, \mathcal{B}, p_{\mathrm{rec}}, D$	veh-eq/s in normalized receiving-work units
f/h	veh-eq/(m s); actual lateral density source
a_r	m/s^2; selected physical recovery acceleration
$\gamma, \delta, \beta, c_j$	dimensionless work coefficients
χ	s^2/(veh-eq m); fixed mismatch coefficient
Θ, t, T	s; duration, time or mass-rank admission time
W	veh-eq; integrated isolated lost service
J	veh-eq s; complete local TST

3 Preparation: conservative transfer limited by adjacent service

3.1 The local rule and free-branch reduction

Own A demand claims receiving before the E offer:

$$S(v,a) = \min\{C(v), w(k_j - a)\}, \quad f = \min\left\{ve, \frac{S(v,a) - va}{\gamma}\right\}, \quad \dot{e} = -f/h, \quad \dot{a} = f/h, \quad \dot{b}_0 = 0. \tag{10}$$

This is H1's opportunity-equivalent closure. Actual transfer over $dx\,dt$ is $(f/h)dx\,dt$; neither f itself nor γf is an additional longitudinal mass flow.

Paired transfer gives $e + a = r$. In the invariant upstream free branch, $S = C$, hence

$$\dot{e} = -\frac{v}{h} \min\left\{e, \frac{k_c(v) - r + e}{\gamma}\right\}, \qquad a = r - e. \tag{11}$$

The scalar right-hand side is Lipschitz. At $e = 0$ it is zero, and the receiving alternative vanishes at $a = k_c(v)$ if that boundary is reached. Thus the initial bounds imply

$$0 \leq e(t) \leq e_0, \quad a_0 \leq a(t) \leq \min\{r, k_c(v)\}, \quad b_0 \leq k_c(v). \tag{12}$$

Under (8), $r < R/v_+ \leq R/v \leq k_c(v)$, with a strict margin inherited from $v_+ r < R$. Uniform source and common motion preserve the remaining body's homogeneity provided the terminal entrance accepts it. Section 5 supplies that protection.

3.2 Progress, explicit branches and the speed effect

Put $z = vt$ and $\Delta(v) = k_c(v) - r > 0$. Then

$$e_z = -\frac{1}{h}\min\{e, (\Delta + e)/\gamma\}. \tag{13}$$

On a donor-limited interval starting at (z_s, e_s),

$$e(z) = e_s e^{-(z-z_s)/h}. \tag{14}$$

On a receiving-limited interval,

$$e(z) = (e_s + \Delta)e^{-(z-z_s)/(\gamma h)} - \Delta. \tag{15}$$

For $\gamma > 1$ the switch is at $e = \Delta/(\gamma - 1)$; the minimum selects the branch. A receiving-limited solution decreases to that value and then follows the donor branch. If $\gamma = 1$, positive Δ makes the donor branch active throughout.

The explicit v factor cancels in progress coordinates. Therefore travel time alone no longer guarantees improved preparation. Since $k_c(v)$ decreases with v, the scalar comparison principle applied to (13) makes $e(z; v)$ nondecreasing with advice at fixed z, allowing equality on donor-only intervals. Speed dependence is through the declared receiving headroom.

4 Finite footprint, handoff arrivals and upstream mass balances

Early transfer stops at the fixed cutoff L_p. No ordinary lateral source acts inside the terminal; residual E merges at L. With zero initial terminal histories, entries satisfy almost everywhere

$$\begin{aligned} &t_f^p = (L_p - \ell)/v, \quad t_r^p = L_p/v, \\ &\dot{F}_E = ve(t), \quad \dot{F}_A = v(r - e(t)), \quad \dot{F}_B = vb_0 \quad (t_f^p < t < t_r^p), \end{aligned} \tag{16}$$

and vanish outside that interval. Point values of flux at endpoints do not affect continuous cumulative mass.

The remaining occupied preparation length is

$$s(t) = \begin{cases} \ell, & 0 \le t \le t_f^p, \\ L_p - vt, & t_f^p < t < t_r^p, \\ 0, & t \ge t_r^p. \end{cases} \qquad J_{\mathrm{lat}}(t) = \int_0^t s(u)f(u)/h \,\mathrm{d}u. \tag{17}$$

Differentiating interval by interval and matching initial values gives

$$\begin{aligned} s\rho_0 + F_E + F_A + F_B &= M, \\ se + F_E + J_{\mathrm{lat}} &= \ell e_0, \\ sa + F_A - J_{\mathrm{lat}} &= \ell a_0, \qquad sb_0 + F_B = \ell b_0. \end{aligned} \tag{18}$$

An early-merged unit belongs to A once; it is not charged as a later forced E merge. The uniform body's density state may be continued algebraically after preparation empties, but no source or population then remains there.

5 Terminal propagation, physical stock and storage protection

For each lane use minimum free travel $\tau = b/v$ and backward delay $\tau_b = b/w$. Negative-time histories are zero. Define

$$P_j(t) = F_j(t-\tau), \quad N_j(t) = F_j(t) - Y_j(t), \quad q_j(t) = P_j(t) - Y_j(t), \quad \lambda_j = \dot{P}_j, \quad g_j = \dot{Y}_j. \tag{19}$$

N_j is physical stock, q_j eligible unserved mass. Feasible departures satisfy $0 \le Y_j \le P_j \le F_j$, so no unit exits before its minimum travel. Queued demand is $C(v)$; an empty free approach offers λ_j. The independent far lane has $Y_B = P_B$ under (8).

For empty-link boundary receiving, the vacancy expression is

$$V_j^{\mathrm{vac}}(t) = \begin{cases} k_j w t - F_j(t), & 0 < t \le \tau_b, \\ Y_j(t-\tau_b) + k_j b - F_j(t), & t > \tau_b. \end{cases} \tag{20}$$

By (12), $\dot{F}_j \le C(v) < w k_j$. Initially $F_j(t) \le C(v)t < k_j w t$. Later $V_j^{\mathrm{vac}} \ge k_j b - M > 0$. Thus terminal entrance receiving never holds preparation in the sufficient domain. This is finite physical storage protection, not an infinite point buffer or a general spillback solution. It also implies $N_j \le M < k_j b$.

The eligible arrival pulse is

$$t_f = (L-\ell)/v, \quad t_r = L/v, \qquad \lambda_A + \lambda_E = vr =: s_v < R, \quad \lambda_A' = -\lambda_E' \in [0, L_A], \quad L_A = v_+^2 e_0/h \tag{21}$$

inside the pulse. E arrivals decrease and A arrivals increase; only the front and rear have external jumps. H1 branch switches preserve continuous rates. This monotonic structure is essential to the later event proof.

6 Current speed marks, immediate work and unique service roots

6.1 Traffic-state closure and the fixed coefficient

Free boundary marks use $u_j = v$; queued marks use $u_j = U(g_j)$. These are equilibrium trace assumptions, not independent speed state or whole-lane mean speeds. The selected W-F work expression is

$$c_A = 1, \qquad c_E = c_0 + \chi g_A [u_A - u_E]_+, \qquad g_A + c_E g_E \le \mathcal{B}. \tag{22}$$

It has no direct advice effect at fixed represented traffic and budget. For queued E, with $a = g_A$, fixed A mark u and available E work x,

$$\mathcal{H}(g; a, u) = g\{c_0 + \chi a [u - U(g)]_+\} = x. \tag{23}$$

On the active mismatch branch,

$$\mathcal{H}_g = c_0 + \chi a\{u - U(g) - gU'(g)\}, \quad U'(g) = \frac{wK}{(K-g)^2}, \quad gU'(g) \le v(1 + v/w) \quad (0 \le g \le C(v)). \tag{24}$$

Because $a \le C(v) \le C_\star$ and $v \le v_\star$, $\mathcal{H}_g \ge \eta > 0$. Outside mismatch, $\mathcal{H}_g = c_0$; the continuous function is strictly increasing across the kink. Also $\mathcal{H}(0) = 0$ and $\mathcal{H}(C(v)) = c_0 C(v) \ge \mathcal{B} \ge x$. There is exactly one feasible root. With the same bounds,

$$c_0 \le c_E \le c_{\max}, \qquad x/c_{\max} \le g_E \le x/c_0, \qquad g_A + g_E \le \mathcal{B} \le R \le C(v). \tag{25}$$

Thus sending/receiving ceilings and mass conservation are respected.

6.2 What fixing the coefficient changed

The earlier trial divided the mismatch term by $C(v)v$. Its multiplier $\rho(v) = (w+v)/(Kv^2)$ has $\rho'(v) = -(v+2w)/(Kv^3) < 0$. When both approaches are queued at fixed budget, A flow/mark are fixed, yet that trial changes E work with an inactive advice cap. W-F removes that direct channel. It agrees with the trial at the fixed anchor v_+; it does not preserve its advice derivative away from that anchor.

On a regular free-A/queued-E branch with fixed $a, \mathcal{B}$, $u_A = v > U(g_E)$ and $x = \mathcal{B} - a$. Implicit differentiation gives

$$\frac{\partial g_E}{\partial v} = -\frac{\chi a g_E}{\mathcal{H}_g} \leq 0. \tag{26}$$

Lowering free A speed can therefore ease E entry in this local comparison. This is not a full-cohort derivative: H1, arrivals, travel and recovery also change. When both are queued below the advice cap, there is no direct v in the local root at fixed budget.

7 Priority, cooperation and causal queue transitions

Free A claims its offer when it fits. Queued A uses equal-work sharing with unused-work redistribution, including when E is free. A convenient fixed-cost template is the minimal $z \geq 0$ with

$$\min(d_A, z) + \min(c_E d_E, z) = \min(\mathcal{B}, d_A + c_E d_E), \quad g_A = \min(d_A, z), \quad g_E = \min(d_E, z/c_E). \tag{27}$$

The selected branches resolve the implicit cost as follows:

Current modes	Service and feasibility
A/E free	$g_A = \lambda_A$, $g_E = \lambda_E$ if $\lambda_A + c_0\lambda_E \leq \mathcal{B}$.
A free, E queued	$g_A = a = \lambda_A \leq \mathcal{B}$, $\mathcal{H}(g_E; a, v) = \mathcal{B} - a$.
A/E queued	$g_A = \mathcal{B}/2$, $\mathcal{H}(g_E; \mathcal{B}/2, U(\mathcal{B}/2)) = \mathcal{B}/2$.
A queued, E free	$g_E = \lambda_E$, $g_A = \mathcal{B} - c_0\lambda_E$ when $c_0\lambda_E \leq \mathcal{B}/2$; otherwise E has a strict offer shortfall and enters its queued branch.

Free E has $u_E = v$ and thus no positive mismatch term for either A mode. With both queued, $g_E \leq g_A/c_0 \leq g_A$; E's maneuver disadvantage persists without a separate arbitrary flow-share parameter.

Causal guards. Initially both queues are zero and modes free. An empty approach remains free while its offer is met. A strict free-mode shortfall selects its queued trace; recomputed right-hand queue growth must be compatible with nonnegative mass. While $q_j > 0$,

$$\dot{q}_j = \lambda_j - g_j, \qquad \dot{Y}_j = g_j. \tag{28}$$

At a true zero hit, retest free admission. Equality alone does not create a queue; use maximal feasible free continuation and right-hand arrival data at jumps. Retain recovery history across mode changes. No numerical epsilon, branch latch, averaging rule or dwell time is added.

Why cooperation cannot be restricted literally to two positive queues: at E clearance with A still queued, take unit costs, budget $4/5$, $\lambda_A = 3/5$, $\lambda_E = 1/10$. Cooperative redistribution gives $(g_A, g_E) = (7/10, 1/10)$, so E stays clear. Strict all-state A priority instead gives E no service at zero but $2/5$ at positive queue, requiring $\dot{q}_E = 1/10$ at zero and $-3/10$ above zero. No nonnegative absolutely continuous continuation satisfies that literal toggle. The selected current-state rule avoids it without allocation memory. This is a local diagnostic, not a new primary initialization.

8 A speed-marked recovery profile and recent-mass renewal

8.1 The isolated boundary deduction

Let $u_R = U(R)$ and $a_r > 0$. Define

$$\begin{aligned} S(\theta; u) &= \sqrt{(w+u)^2 + 2wa_r\theta}, \quad V = S - w, \\ Q_{\text{iso}}(\theta; u) &= K\left[\theta - \frac{S-w-u}{a_r}\right], \quad \partial_\theta Q_{\text{iso}} = K(1 - w/S) = c(V), \\ D(\theta; u) &= [R - c(V(\theta; u))]_+. \end{aligned} \tag{29}$$

This reuses the isolated accelerated-insertion point expression to form a capped boundary service loss. V is not a vehicle's speed trajectory; it parametrizes the boundary passing rate. The mark u is the selected representative insertion speed. The source isolated deduction does not validate the aggregate overlap rule below.

For $u < u_R$,

$$\begin{aligned} \Theta(u) &= \frac{(w+u_R)^2 - (w+u)^2}{2wa_r}, \quad \Theta_{\max} = \Theta(0) = u_R/a_r + u_R^2/(2wa_r), \\ W(u) &= \int_0^\infty D(\theta; u)\,\mathrm{d}\theta = \frac{K}{w+u_R}\frac{(u_R-u)^2}{2a_r}. \end{aligned} \tag{30}$$

For $u \geq u_R$, D, W, Θ are zero. To obtain the area, change variable from age to V:

$$\frac{d\theta}{dV} = \frac{w+V}{wa_r}, \qquad D\frac{d\theta}{dV} = \frac{(K-R)(u_R-V)}{wa_r}, \qquad \frac{K-R}{w} = \frac{K}{w+u_R}. \tag{31}$$

Integrating from u to u_R gives (30). On the active part,

$$D_\theta = -\frac{Kw^2a_r}{S^3}, \quad D_u = -\frac{Kw(w+u)}{S^3}, \quad D_{a_r} = -\frac{Kw^2\theta}{S^3}. \tag{32}$$

Thus D decreases with age and mark, $0 \leq D \leq R$, $|D_\theta| \leq a_r k_j$ and $|D_u| \leq k_j$. The bounds extend through the continuous zero-loss cutoff. Physical age is not a numerical step. W is a lost-service area, not a second immediate E charge.

8.2 The selected E-only renewal account

For $Y = Y_E$, let $T(\xi)$ be its mass-rank admission-time inverse and $a_m(t) = [Y(t) - m]_+$. The selected account is

$$p_{\text{rec}}(t) = \frac{1}{m}\int_{a_m(t)}^{Y(t)} D(t - T(\xi); u_E(T(\xi)))\,\mathrm{d}\xi, \qquad \mathcal{B}(t) = [R - p_{\text{rec}}(t)]_+. \tag{33}$$

Only E creates new loss. A influences E service/marks and consumes the shared budget; A's slow passage is not independently charged as a new profile. The account averages at most the latest mass m. New mass retires equal oldest weight after fill while all profiles age. Recent speed/age history, not just the reading, is retained predicted state.

Flat intervals of Y carry no admission measure; inverse choices there do not affect the result. Equivalently,

$$p_{\text{rec}}(t) = \frac{1}{m}\int_0^t D(t-s; u_E(s))g_E(s)\mathbf{1}_{\{Y(s) > a_m(t)\}}\,\mathrm{d}s. \tag{34}$$

Only ages up to $\Theta_{\max}$ and the latest mass m can contribute. Where derivatives exist, put $T_o = T(Y(t) - m)$ when $Y(t) > m$. Then

$$\dot{p_{\rm rec}} = \frac{g_E(t)}{m} D(0; u_E(t)) + \frac{1}{m} \int_{a_m(t)}^{Y(t)} D_\theta(t - T(\xi); u_E(T(\xi)))\, \mathrm{d}\xi$$
$$- \frac{g_E(t)}{m} \mathbf{1}_{\{Y(t)>m\}} D(t - T_o; u_E(T_o)). \tag{35}$$

These are injection, aging and oldest-weight retirement. There is no division by an old flow rate. At fill and other endpoints use the a.e. identity; the cumulative account is continuous. With $g_E = 0$ only aging remains. A fast new E entry can create zero loss and still retire an older strong profile; this is a selected approximation requiring physical evidence.

9 Local causal construction and derived positive service

9.1 The moving cutoff does not require a positive old rate

Compare two histories (g_i, u_i, Y_i) with identical past through t_0, $0 \le g_i \le G := R/c_0$ and bounded marks. For fixed $t > t_0$, put

$$\epsilon = \sup_{s \le t} |Y_1(s) - Y_2(s)| \le \int_{t_0}^{t} |g_1 - g_2|\, \mathrm{d}s, \qquad |a_{m,1}(t) - a_{m,2}(t)| \le \epsilon. \tag{36}$$

In (34), first hold one indicator fixed. Bounded D and its mark Lipschitz constant give $R \int |\Delta g| + k_j G \int |\Delta u|$. Where the indicators differ, $|Y_2(s) - a_{m,2}(t)| \le 2\epsilon$. The $g_2\, ds$ mass in this rank band is at most 4ϵ, including any zero-rate plateaus. Hence

$$|\Delta p_{\rm rec}(t)| \le \frac{1}{m} \left(5R \int_{t_0}^{t} |\Delta g_E|\, \mathrm{d}s + k_j G \int_{t_0}^{t} |\Delta u_E|\, \mathrm{d}s \right). \tag{37}$$

The argument also covers recorded old mark jumps. It avoids differentiating a rank inverse or assuming an unphysical lower bound on E flow.

On a valid selected node regime, (24) and bounded U' on $[0, R]$ give local uniform Lipschitz constants L_g, L_u for the E flow and its mark as functions of $\mathcal{B}$. The budget map is 1-Lipschitz. On a sufficiently short interval with the same past,

$$\|\Delta \mathcal{T}\mathcal{B}\|_\infty \le \frac{\tau}{m}(5RL_g + k_j G L_u) \|\Delta \mathcal{B}\|_\infty. \tag{38}$$

Choose the proof interval short enough for contraction and to remain in a valid chart. Bounded increments give causal construction and continuous dependence until its next guard. This is a proof device, not a simulator time step. It does not by itself establish the validity or uniqueness of every queue switch; Section 10 joins the selected regimes.

9.2 Positive budget is a consequence, not an added floor

Admitted E rate obeys $g_E \le G$. Since $D(\theta; u) \le D(\theta; 0)$ decreases with age, the maximal possible account fills the youngest ages at rate G up to mass m. Rearrangement gives

$$p_{\rm rec} \le p_{\max} := \frac{G}{m} \int_0^{m/G} D(\theta; 0)\, \mathrm{d}\theta < R, \qquad \mathcal{B} \ge B_{\min} := R - p_{\max} > 0. \tag{39}$$

Strictness follows from $a_r > 0$, making $D(\theta; 0) < R$ for positive age. The bound also holds before the window fills. It is common across advice at fixed parameters, but can be very small. Clipping in (33) is inactive in this domain.

From (35), the retained mass is at most m, each boundary term is bounded by GR/m and $|D_\theta| \le a_r k_j$. Therefore the account is Lipschitz with

$$|\dot{\mathcal{B}}| \le L_B := 2GR/m + a_r k_j \quad \text{a.e.} \tag{40}$$

This allows mark jumps at queue events. Also the renewal integrand is a subset of the nonnegative all-history additive integrand, so Tonelli gives

$$\int_0^\infty p_{\mathrm{rec}}(t)\,\mathrm{d}t \le \overline{M}_E W_0/m, \quad \overline{M}_E = P_E(L/v) \le M, \quad W_0 = W(0). \tag{41}$$

It is a bound, not adoption of the additive rival.

9.3 Two retained limits

For a constant free A offer $a < R$, $g_E \le (\mathcal{B} - a)/c_0$ and (35) gives $\dot{p_{\mathrm{rec}}} \le R g_E/m$. With $x = \mathcal{B} - a$, $\dot{x} \ge -Rx/(mc_0)$, hence

$$x(t) \ge (R - a)e^{-Rt/(mc_0)} > 0. \tag{42}$$

The selected smooth E source cannot generate a first A queue in finite time in that constant-A limit. H1's A arrivals increase, adding $-\dot{a}$ to $\dot{x}$; Section 13 constructs A delay there.

For a constant E rate and mark across a filled renewal window,

$$p_{\mathrm{rec}} = \frac{g_E}{m} W_{m/g_E}(u_E), \quad W_h(u) = \int_0^h D(\theta; u)\,\mathrm{d}\theta, \quad g_A + c_E g_E = R - \frac{g_E}{m} W_{m/g_E}(u_E). \tag{43}$$

This is a restricted stationary target for a simpler instantaneous rival, not an exact transient reduction or proof that history is necessary.

10 Whole-domain events, unique continuation and finite clearance

The result below is restricted to (8)–(9), the monotone H1 pulse, neutral initial account and selected causal guards. It is not a general invariant LTM junction theorem.

10.1 E has at most one waiting episode in this primary domain

Initially $\mathcal{B} = R$ and A's offer is below R. Let

$$W_f(v) = \lambda_A(t_f+) + c_0\lambda_E(t_f+). \tag{44}$$

If $W_f \le R$, both free streams pass. Because $R \le C(v_-)$, $v \ge u_R$, so free E has zero new loss. The free work $s_v + \delta\lambda_E$ decreases through the pulse; free passage persists. If $W_f > R$, E's free service test fails and the queued root satisfies $g_E \le (R - \lambda_A)/c_0 < \lambda_E$. This is a compatible strict E birth while A remains free.

At a later first E zero hit during the pulse, if A is free, the pre-hit queued rate implies

$$\lambda_E \le g_E^Q \quad \Longrightarrow \quad \lambda_A + c_0\lambda_E \le \mathcal{B}. \tag{45}$$

If A is queued, its half-work branch implies

$$c_0\lambda_E \le c_0 g_E^Q \le \mathcal{B}/2. \tag{46}$$

Thus free E is feasible. Its new profiles are zero; old profiles only age or retire, making $\mathcal{B}$ nondecreasing. E's arrival offer keeps decreasing, so (46) persists while A queues. At A clearance its former service $\mathcal{B} - c_0\lambda_E$ admits joint free flow, and the joint free workload thereafter decreases. Simultaneous clearance also gives joint feasibility because both individual half-work offers fit. After the pulse rear there are no new arrivals. E therefore cannot re-form after clearing in this domain.

Scope is essential. The proof uses the monotone H1 pulse, free-mark neutrality and selected recovery/priority law, not finiteness alone. Later/bursty E arrivals in another setting or native finite cohort can form another queue; that event is not automatically a failed evaluator.

10.2 A events cannot accumulate infinitely fast

While E queues during the pulse, A is born when its free offer first loses feasibility at $\lambda_A = \mathcal{B}$. Switching to cooperation gives $\dot{q}_A = \mathcal{B}/2 \geq B_{\min}/2 > 0$. If A later clears before E, the pre-hit derivative requires $\lambda_A \leq \mathcal{B}/2$. For $y = \mathcal{B} - 2\lambda_A$, birth has $y = -\mathcal{B} \leq -B_{\min}$ and clearance has $y \geq 0$. By (21) and (40), a birth-to-clearance interval is at least

$$B_{\min}/(L_B + 2L_A). \tag{47}$$

At clearance the free margin $x = \mathcal{B} - \lambda_A \geq B_{\min}/2$. A later birth needs $x = 0$, taking at least $B_{\min}/[2(L_B + L_A)]$. A common smaller separation is

$$\tau_0 = \frac{B_{\min}}{2(L_B + 2L_A)} > 0. \tag{48}$$

It is derived, not imposed as a dwell timer. E clearance and the single pulse rear add finitely many exceptions. Once E is free, a cleared A cannot re-form because joint free work fits and decreases while service recovers. After the rear, positive queues only drain. Tangential zero hits use the same free-admission test and still have the post-clearance margin.

10.3 Joining the regimes and clearing the cohort

Theorem 10.1 (Conditional complete forward evolution)**.** *For every constant advice in the original interval, under the retained primary assumptions and causal convention, the selected law has a unique causal piecewise-regime continuation through full cohort clearance.*

Proof. Use the local contraction in Section 9 on each valid chart. At the front E birth has the strict compatible sign above. An A birth has positive right-hand derivative. At A zero hits free admission is feasible; at E zero hits use (45)–(46). Simultaneous hits admit both free streams. Equality alone retains maximal feasible free continuation, so an alternative queued root cannot create an untriggered queue. Flux values at individual switching instants do not alter cumulative mass.

State and service remain bounded. The finite pulse admits at most one E episode, and (48) prevents infinitely many A switches on it. After the rear no new queues arise, with at most two clearances. Thus local continuation joins into one unique causal history. No transversality condition or reduced advice neighborhood is imposed for this forward result. Smooth advice dependence is a separate claim.

All mass is eligible by $t_r = L/v$. If $Q = q_A + q_E > 0$ thereafter, available work is used, so $g_A + g_E \geq \mathcal{B}/c_{\max} \geq B_{\min}/c_{\max}$. Consequently

$$T_{\text{clear}}(v) \leq \frac{L}{v} + \frac{c_{\max} Q(L/v)}{B_{\min}} \leq T_{\max} := \frac{L}{v_-} + \frac{c_{\max} M}{B_{\min}} < \infty. \tag{49}$$

The far lane clears without a queue. Remaining profiles then age out within their finite support. This finishes the forward construction. □

Another sufficient clearance bound follows by integrating current work and (41) after the rear:

$$T_{\text{clear}} \le \frac{L}{v} + \frac{c_{\max} Q(L/v) + \overline{M}_E W_0/m}{R}. \tag{50}$$

Neither estimate asserts a practical simulation runtime or physical realism of the sufficient parameter domain.

11 Conservation and the complete local objective

Combining (18) and (19) gives

$$s(t)\rho_0 + \sum_j N_j(t) + \sum_j Y_j(t) = M. \tag{51}$$

All quantities are continuous traffic mass. Work weights change available service, never the amount transferred or removed. Retiring a profile removes information weight, not population.

Full local TST is

$$J(v) = \int_0^{T_{\max}} \left[M - \sum_j Y_j(t) \right] \mathrm{d}t. \tag{52}$$

By (49), using any larger bound adds only zero. Every initial position $x \in [0, \ell]$ has no-delay eligibility time $(L - x)/v$. Paired early transfers do not change the total eligibility history. Hence

$$\int_0^\infty \left[M - \sum_j P_j(t) \right] \mathrm{d}t = \int_0^\ell \rho_0 \frac{L - x}{v} \,\mathrm{d}x = \frac{M(L - \ell/2)}{v},$$

$$M - \sum_j Y_j = M - \sum_j P_j + \sum_j q_j,$$

$$J(v) = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E) \,\mathrm{d}t. \tag{53}$$

The far lane has $q_B = 0$. Both A and E waiting are included exactly once. Eligible waiting area measures excess time; it is not a physical queue length or a declaration of when a vehicle first reaches a queue tail.

Completion is at the choke. A crossed E unit can retain a causal restriction for following unfinished traffic, whose waiting is counted. The completed unit's own post-line travel is excluded. Adding a recovery-zone objective would be a separate G4 choice; it is not adopted by this accounting.

12 Initial information, negative controls and sensitivity limits

Known initial E mass E_0 and continuing symmetry reconstruct

$$e_0 = E_0/\ell, \qquad a_0 = b_0 = (M - E_0)/(2\ell). \tag{54}$$

Together with fixed parameters and empty terminal/account histories, the forward theorem yields identical predictions for aggregate-plus-E_0 and explicit lane state at the same information, for every permitted advice. Predicted recovery history is not an extra initial measurement; the current meter reading alone is not sufficient dynamic state.

The archived predecision aggregate-history collision remains upstream of an empty terminal and is unaffected by the terminal substitution. Its reachability/observation assumptions remain conditional. No action split follows solely from that ambiguity: the amended objectives still require

their own comparison. Strong aggregate/history, scalar, model-form and bounded-search rivals remain in scope.

If $\delta = 0$, the front free work is $s_v < R$. Equal free marks make the mismatch term zero; no queue or recovery activates. Therefore

$$J(v) = M(L - \ell/2)/v \tag{55}$$

and fastest advice is common for every preparation. No E is another neutral limit. This is a property of the selected ordinary-underload member, not a universal explanation of congestion.

With $z_f = L_p - \ell$ and $e_f(v) = e(z_f; v)$, H1 comparison makes e_f nondecreasing in v. Thus

$$W_f(v) = v\{r + \delta e_f(v)\} \tag{56}$$

is strictly increasing for $r > 0$. There is at most one initial queue threshold; it can lie outside the advice interval. Equality retains free passage. At an interior threshold with $a = \lambda_A > 0$, $e = \lambda_E > 0$, $a + c_0 e = R$ and $\chi > 0$,

$$\mathcal{H}(e; a, v) - (R - a) = \chi a e\{v - U(e)\} > 0. \tag{57}$$

The queued-root limit is below the free E flow, so local flux/marks may change abruptly on activation. This is not a proof that full J jumps. Forward uniqueness does not establish smooth derivatives across it.

Within regular charts, (26) and ordinary implicit derivatives describe local response. A full-objective derivative must also include H1 branch changes, action-dependent arrival times, recovery marks/aging/retirement and queue events through complete clearance. This report does not execute or finish that remaining gradient review; it records why S13's simpler derivative cannot be reused. No AD, numerical finite differences, policy optimization or useful-gradient claim follows.

13 A complete analytical family with A delay and active renewal

This consolidates the OR-Q1 construction with W-F at its fixed anchor $v_0 = v_+$. There the old trial and selected work expressions coincide exactly. The result is coexistence under the retained equations, not a calibrated scenario, numerical rollout, chosen task setting or optimal- advice certificate. Long geometry and very slow recovery may be needed.

13.1 H1 input and finite-prefix auxiliary limit

Translate the eligible front to time zero. A donor-limited H1 pulse at v_0 has arrival rates

$$e(t) = e_f e^{-kt}, \quad a(t) = s - e(t), \quad k = v_0/h > 0, \quad s = v_0 r < R, \quad \varepsilon = R - s > 0. \tag{58}$$

Only in this construction, e, a denote arrival rates, not densities; e_f/v_0 is the corresponding front density. Choose $c_0 > 1$, $e_f > 0$, $a_f = s - e_f > 0$ with

$$\varepsilon < \delta e_f/2, \quad G_0 = (\varepsilon + e_f)/c_0 < R, \quad d_0 = R - G_0 > \varepsilon + e_f, \quad \delta = c_0 - 1. \tag{59}$$

The initial free work is $a_f + c_0 e_f = R - \varepsilon + \delta e_f > R$. E therefore queues first under the selected priority.

Temporarily set $a_r = \beta = 0$ solely as an auxiliary finite-prefix limit. Before window fill, while A is free and E queued,

$$g_E = \frac{\mathcal{B} - a(t)}{c_0}, \quad \dot{\mathcal{B}} = -\frac{g_E(R - g_E)}{m}, \quad \dot{q}_E = e - g_E. \tag{60}$$

This follows from the zero-aging profile and $D(0; U(g_E)) = R - g_E$. Since $g_E \le G_0$ and $\mathcal{B} \ge a \ge a_f$,

$$R - \mathcal{B} \ge d_0 Y_E/m, \qquad Y_E \le m(\varepsilon + e_f)/d_0 < m. \tag{61}$$

A first fill before A onset would contradict this bound, so the unfilled prefix is self-consistent.

While $\mathcal{B} \ge s$ and $e \ge e_f/2$,

$$-\dot{\mathcal{B}} \ge \mu := \frac{e_f d_0}{2mc_0} > 0, \qquad \dot{q}_E \ge (\delta e_f/2 - \varepsilon)/c_0 > 0. \tag{62}$$

Take positive k sufficiently small that $k\varepsilon/\mu < \log 2$. The budget reaches s by ε/μ while the condition on e holds, then passes below s. Afterwards, while A is still free, $g_E < e/c_0 < e$, preventing E clearance. Because $a(t) \uparrow s$ and the budget is nonincreasing, a finite first A onset t_A occurs. Its pre-hit margin $x = \mathcal{B} - a$ has

$$x(t_A) = 0, \quad \dot{x}(t_A-) = -ke(t_A) < 0, \qquad \dot{q}_A(t_A+) = \mathcal{B}(t_A)/2 > 0. \tag{63}$$

Thus increased A arrivals can cross the protected-service margin even though the constant-A limit in (42) cannot.

13.2 A backlog large enough to exercise renewal

At $k = 0$ in the same auxiliary input, A is constant a_f. The finite-time barrier keeps $\mathcal{B} > a_f$, $Y_E < m$ and E queued, with

$$\dot{q}_E \ge \sigma := (\delta e_f - \varepsilon)/c_0 > 0. \tag{64}$$

Choose finite T_b with $\sigma T_b > 3m$. The free-A margin remains strict on that compact interval. Continuous dependence of (60) on its input permits a sufficiently small positive k for A to remain free through T_b with $q_E(T_b) > 2m$. Retain (62). The finite A onset is later and E cannot clear in between. This removes constant input as a primary assumption; $k = 0$ is only a finite-prefix comparison.

End the actual finite pulse at $T > t_A$ sufficiently close to t_A that $q_A(T) > 0$, $q_E(T) > 2m$, $Y_E(T) < m$, $\mathcal{B}(T) > 0$, and

$$\frac{q_A(T)}{c_0} < m - Y_E(T), \quad q_A(T) < c_0 q_E(T), \quad \frac{R q_A(T)}{mc_0} < \mathcal{B}(T)/2. \tag{65}$$

The first A queue tends to zero as $T \downarrow t_A$, while the other margins stay positive, so these inequalities are attainable.

In the auxiliary post-pulse continuation, both queues drain with $g_A = \mathcal{B}/2$, $g_E = \mathcal{B}/(2c_0)$ before fill. A clears first, using $q_A(T)/c_0$ E mass. Since $d\mathcal{B}/dY_E = -(R - g_E)/m \ge -R/m$, (65) keeps $\mathcal{B} \ge \mathcal{B}(T)/2$ until this strict clearance, before fill. Afterwards A has no arrivals and $g_E = \mathcal{B}/c_0$.

The first $Y_E = m$ is reached in finite time with E still queued. Indeed, before it,

$$\mathcal{B} = R(1 - Y_E/m) + \frac{1}{m}\int_0^{Y_E} g_E(T(\xi))\,\mathrm{d}\xi. \tag{66}$$

If Y_E only approached m asymptotically, the positive earlier integral would give a positive limiting budget and hence positive limiting service, a contradiction. At fill, $g_E > 0$ and the oldest profile has positive loss. Thus the retirement term in (35) is nonzero immediately afterwards. The construction exercises genuine renewal, not only its initial additive prefix.

13.3 Removing the auxiliary zero parameters

On a fixed finite interval the positive-recovery kernel converges uniformly to $[R - c(u)]_+$ as $a_r \downarrow 0$, with error at most $a_r k_j$ times interval length. The root is uniformly regular for sufficiently small β satisfying (9). The Volterra estimates give finite-prefix continuous dependence on these parameters in each valid regime.

The constructed sequence has strict margins or transverse hits: E front shortfall, A onset (63), pulse end with positive queues, A clearance with positive service, and fill with $g_E > 0$. Local continuous dependence and the nonzero guard derivatives preserve the finite sequence and its inequalities for sufficiently small *positive* a_r and *positive* β. A has positive waiting area and oldest-profile retirement remains nontrivial. The auxiliary zero-recovery limit is not part of the admitted model; its infinite-time behavior is not used.

For orientation, an all-history additive O-S account started identically agrees with O-R until first fill t_R. At that common state only O-R retires old weight, so

$$(\dot{\mathcal{B}}_{O\text{-}R} - \dot{\mathcal{B}}_{O\text{-}S})(t_R+) = \frac{g_E(t_R)}{m} D(t_R; u_E(0+)) > 0. \tag{67}$$

This is a local difference, not global policy ordering or evidence that one closure is physically correct. O-S remains unselected.

After the pulse, no new queues are born. At most two clearances remain; the positive-budget/clearance bounds make them finite. The small A tail preserves A-first clearance, then E alone clears. This gives a complete unique regular family with positive A delay and active renewal.

13.4 Realizing it with the unchanged H1 geometry

Choose $r = s/v_0$, $e_0 + a_0 = r$, $a_0 = b_0 > 0$. A receiving-limited initial branch followed by a donor-limited front is possible when

$$k_c(v_0) - r < (\gamma - 1)e_0, \qquad e_f/v_0 < \frac{k_c(v_0) - r}{\gamma - 1}. \tag{68}$$

Choose front progress z_f after the switch. Increasing h at fixed z_f/h retains that density while making $k = v_0/h$ small. Put

$$\ell = v_0 T, \quad L_p = \ell + z_f, \quad M = \ell(r + b_0), \quad b > M/k_j, \quad L = L_p + b. \tag{69}$$

The terminal length changes only the common translated eligibility delay, not the node argument. All original advice-domain and far-lane conditions can be retained; no advice subinterval is silently substituted.

An exact normalized static feasibility point from the earlier proof is

$$R = m = 1, \quad c_0 = 11/10, \quad e_f = 1/8, \quad \varepsilon = 1/512, \quad s = 511/512, \tag{70}$$

giving

$$\begin{gathered} d_0 = 2491/2816, \quad \mu = 12455/247808, \quad \varepsilon/\mu = 484/12455, \\ \sigma = 27/2816, \qquad (\varepsilon + e_f)/d_0 = 715/4982 < 1. \end{gathered} \tag{71}$$

One compatible H1 choice uses

$$\begin{gathered} v_- = 1, \quad v_0 = v_+ = 2, \quad w = 2, \quad k_j = 3, \\ a_0 = b_0 = 1/8, \quad e_0 = 383/1024, \quad r = 511/1024, \quad \gamma = 5, \quad R_B = 1. \end{gathered} \tag{72}$$

Its fast front density $1/16$ is reached at

$$z_f/h = 5\log(5632/5125) + \log(1025/256). \tag{73}$$

These are static algebraic existence checks, not selected traffic parameters. k, terminal geometry and positive recovery/mismatch values are chosen by strict finite-prefix arguments, not by a numerical rollout. The witness persists locally near the anchor; the separate forward theorem qualifies all original advices, without claiming A delay at all of them. No preferred advice, interior optimum or useful gradient is established.

14 What physical observations can and cannot identify

This collects MP6-Q1's conditional algebra; it neither executes nor opens calibration. The required informative observations may not be available in the primary cohort. A future execution proposal must specify their scale, uncertainty, tolerances and development/held-out separation.

14.1 Recovery tails and immediate work

If no more E is entering and A remains queued, $g_E = 0$, $c_A = 1$, and work conservation gives

$$g_A = \mathcal{B} = R - p_{\mathrm{rec}}, \qquad p_{\mathrm{rec}} = R - g_A. \tag{74}$$

With independently known R, the ideal saturated tail exposes the account without current E maneuver work. A free A tail does not: low outflow may simply be low demand. One tail need not identify every profile parameter or test overlapping-history retirement.

At a binding mixed observation with $g_E > 0$, set

$$X = g_A[u_A - u_E]_+, \qquad Z = \frac{R - p_{\mathrm{rec}} - g_A}{g_E} - 1 = \delta + \chi X. \tag{75}$$

Once recovery and marks are separately constrained, two distinct X values give

$$\chi = \frac{Z_2 - Z_1}{X_2 - X_1}, \qquad \delta = \frac{X_2 Z_1 - X_1 Z_2}{X_2 - X_1}. \tag{76}$$

The design determinant is $X_2 - X_1$. Equal covariates do not separate the coefficients; all $X = 0$ leaves χ unresolved. Nonbinding service gives only a feasibility inequality. Admissibility still requires the nonnegative and root-domain bounds, not merely a fitted equality.

Without independent recovery information, a single observation has the accounting ambiguity

$$\delta' = \delta + \epsilon, \quad p'_{\mathrm{rec}} = p_{\mathrm{rec}} - \epsilon g_E, \quad p'_{\mathrm{rec}} + g_A + (1 + \delta' + \chi X) g_E = p_{\mathrm{rec}} + g_A + (1 + \delta + \chi X) g_E. \tag{77}$$

This is not an alternative O-R solution or a global identifiability theorem: the primed account need not satisfy the history law. It shows why a total discharge/TST fit alone cannot establish that present work and retained restriction describe independent real effects. Adding W again to c_E would double-charge the modeled profile loss. Distinct empirical attribution still requires tails, mixtures and joint predictions with one parameter set.

14.2 Preparation and stationary longitudinal correspondence

From (14)–(15), informative pure branches give

$$-\frac{d \log e}{dz} = 1/h \quad \text{(donor)}, \qquad -\frac{d \log(e + \Delta)}{dz} = 1/(\gamma h) \quad \text{(receiving)}. \tag{78}$$

The receiving branch alone primarily identifies the product. The transformation $h' = sh$, $\gamma' = \gamma/s$ preserves it provided the branch and domain remain unchanged. A donor branch can constrain h separately; a final E count generally cannot identify the full law.

For a documented ideal constant-speed headway relation with standing spacing d and desired time gap τ_d,

$$k_{\text{ideal}}(s) = \frac{1}{d + \tau_d s}, \qquad f_{\text{ideal}}(s) = \frac{s}{d + \tau_d s}. \tag{79}$$

Setting $k_j = 1/d$ and $w = d/\tau_d$ makes $k_c(s) = k_{\text{ideal}}(s)$ and $c(s) = f_{\text{ideal}}(s)$. This is a stationary curve identity at actual speed s. It establishes neither native transient wave speed nor instantaneous advice compliance, and it does not prove the stream-specific queued mark $U(g_j)$ at a merge.

14.3 Joint physical admission remains unresolved

One common physical parameter set must explain upstream transfer, actual crossing-speed marks, mixed discharge and recovery while satisfying the full-advice primary domain. Actual acceleration constrains the source meaning of a_r; discharge tests its boundary effect. V must not be fitted as a vehicle-speed trajectory. The physical renewal normalization is fixed; a different window would require reopening its contract.

Development trajectory observations are evidence for common parameters, not extra future inputs to a controller or prescribed native outcomes. Fit neither preferred actions nor gradient signs, and do not select only interior-optimum cases. The mathematical witness and illustration values from source papers are not a jointly supported applied range. No theory-only primary is adopted; that would need an explicit role decision.

15 Provenance, artifact map and remaining scientific boundary

The following source records were inspected for consolidation. They retain their original equations and delivery-time statuses. This report assembles only the selected current member and labels historical results explicitly.

Mathematical responsibility	Archived record and use here
Original S13	Model contract and certificate analysis; Section 1, unchanged historical reference.
Preparation, footprint, finite terminal	H1/T1 joined contract; Sections 2–5. Its old strict-priority one-queue server is superseded, not included as current service.
Root, priority and clearance convention	S-E audit and S-E-P/recovery audit; Sections 6–7, with subsequently selected W-F/R-E/O-R and global qualification.
Recovery update, causal bound and witness	O-R qualification; Sections 8, 9, 13. Its advice-normalized trial work is replaced by W-F; the witness is preserved at the exact anchor.
Fixed coefficient and advice causality	J4 speed-response audit and W-F selection; Section 6. W-A is unselected.
Whole original domain, events, clearance	WF-Q1 forward qualification; Sections 10–12.
Physical interpretation and identification	MP6-Q1 note and joint admission contract; Section 14. Applied admission remains open.

The already reviewed primary literature motivates separate ingredients: lane conservation and mandatory transfer; receiving-work and allocation; boundary-count propagation; and bounded-acceleration point insertion. The exact homogeneous preparation, stream speed/work closure, causal priority and recent-mass renewal remain our adaptations. Their proof arguments are not attributed to a generic CTM/LTM theorem or a published complete lane-drop model.

Detailed archived source and decision map

What is complete. All conditional forward equations, initialization, units, selected mode rules, history update, conservation and objective are specified. Existing mathematical arguments give unique continuation and finite clearance over the retained original advice interval, as well as a nonempty analytical coupled-delay family. The symbolic verification checks supporting identities and rational static inequalities only.

What is not established. No calibrated joint regime, useful action split, full-lane/discrete-order necessity, AD correctness, empirical recovery law or external policy advantage is claimed. Local activation-flux gaps do not prove a full-objective jump. Native finite-cohort requeueing is not automatically invalid. Strong aggregate, scalar and search rivals remain.

Feedback 8 is now answered by consolidation, not by executing the next protocol block. The active checkpoint remains MP6-Q1. Applied MP6 and Step 3 stay open; Step 4 and simulator/experiment work remain unopened.